

04 — DP Redundancy 400 kW (the RESERVE function)

Proves · RESERVE ≠ PEAK SHAVING, and two operational modes inside one calculation.

Mechanics this scenario turns on

- A **Reserve** row is not a swing: **H** = the full requirement, **CoverageFactor** = 1.00 (kW for kW), and it counts toward Spinning Reserve but **not** toward Peak Shaving. A covered reserve is readiness, so it never appears in the demand.
- Every active mode runs its **own** Level 1 — own demand, own combinations, own baseline, own optimum, own t/h. There is no single "tonnes per hour" for the vessel; the year is $\Sigma (\text{mode t/h} \times \text{mode hours})$.
- Level 2 and Level 3 are computed for **Transit only** (decision D4/Q5 — no workbook counterpart elsewhere). Other modes get an empty result, and the pinned-baseline radio does not reach them.
- The **Assumed Configuration** table shows Transit's combinations only, in t/h. The Fuel Consumption figure above it is the annual total across **all** modes.
- Savings are a **difference**, so a mode whose baseline equals its optimum contributes zero — it is present on both sides and cancels, not excluded.
- **Math.Max(0, ...)** bites on short lists: with only two combinations, **2 - 3** clamps to **0** — the optimum itself — so that mode reports **zero** Level 1 savings. A battery in a mode can therefore suppress that mode's Level 1 savings; the value moves to the Battery Benefit badge instead.
- With several modes the Power Demands tables gain one row per mode, and the header is an **hours-weighted average** (total energy ÷ total hours), not a sum.

Panels described below · Battery Contribution · Power Demands · Baseline — two modes, two different rules · IL1 · Battery Benefit

Anything not described here behaves exactly as in **01-excel-baseline** — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 03.

Two modes: Transit 5 000 h (11 463/3 800, NO battery there) and DP 2 000 h (thrust 2 500, hotel 1 500, redundancy 400). Battery 500 kW, **DP only**.

Battery Contribution (DP cascade, budget 500)

Row	H	I	J	L	Note
DpReserve (RESERVE, D=1.0)	400 (the FULL requirement)	400	400	0	covered kW-for-kW, zero factor loss
DpDemand (D=0.5, vf=0)	2 500×0 = 0	0	0	0	thrusters have no swing in the model
Hotel (D=0.05)	1 500×2 % = 30	30	1.5	28.5	budget left: 70

Tiles: **SR = 28.5 · PS = 1.5** (PS counts peak-shaving rows only; the reserve's 400 is full coverage, not a ± band — Excel's J10 would show 401.5).

Power Demands (two rows per table now)

- Transit: propulsion **11 463** (clean!), hotel 3 800 = SG 3 250 + AE **550**
- DP: propulsion **2 500**, hotel 1 500+28.5 = **1 529** — all via SG (the 400 is NOT in the demand: a reserve is readiness, not load)
- Energy: Transit ME $(11\,463 + 3\,250) \times 5\,000 = \mathbf{73\,565\,000}$ · DP ME $(2\,500 + 1\,528.5) \times 2\,000 = \mathbf{8\,057\,000}$ · AE $550 \times 5\,000 = \mathbf{2\,750\,000}$
- Header = hours-weighted: ME $(73\,565\,000 + 8\,057\,000) / 7\,000 = \mathbf{11\,660}$ · AE $2\,750\,000 / 7\,000 = \mathbf{393}$

Baseline — two modes, two different rules

- Transit (no battery there) → default = last row: 2 ME+SG, 2.62547 → 13 127.3
- DP (battery active) → 3rd-from-worst clamped: only 2 combos ⇒ index 0 = its own optimal, $0.703744 \times 2\,000 = 1\,407.5$
- Baseline total = **14 534.8 t/yr**

IL1

Transit optimal $2.578548 \times 5\,000 = 12\,892.7$ + DP 1 407.5 = **14 300.2 t/yr**. Savings = **234.6 t = \$182 996** — ALL from Transit (DP baseline = DP optimal after the clamp).

Battery Benefit

DP only: the twin must spin 430 kW of reserve (400+30) vs our 28.5. Benefit = $\Delta\text{FOC} \times 2\,000\text{ h} = \mathbf{139.9\text{ t/yr} = \$109\,113}$.

Takeaway: RESERVE ≠ PEAK SHAVING. A reserve wants the full amount, is covered 100 %, adds nothing to demand when covered — the battery's strongest use-case (a class requirement satisfied with zero extra spinning iron).